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Efficiency of poll-based multi-period forecasting systems for German state elections ☆

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ABSTRACT

Election polls are frequently employed to reflect voter sentiment with respect to a particular election (or fixed-event). Despite their widespread use as forecasts and inputs for predictive algorithms, there is substantial uncertainty regarding their efficiency. This uncertainty is amplified by judgment in the form of pollsters applying unpublished weighting schemes to ensure the representativeness of the sampled voters for the underlying population. Efficient forecasting systems incorporate past information instantly, which renders a given fixed-event unpredictable based on past information. This results in all sequential adjustments of the fixed-event forecasts across adjacent time periods (or forecast revisions) being martingale differences. This paper illustrates the theoretical conditions related to weak efficiency of fixed-event forecasting systems based on traditional least squares loss and asymmetrically weighted least absolute deviations (or quantile) loss. Weak efficiency of poll-based multi-period forecasting systems for all German federal state elections since the year 2000 is investigated. The inefficiency of almost all considered forecasting systems is documented and alternative explanations for the findings are discussed.

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1. Introduction

Fixed-event forecasts target a particular event at a specific future date and can only be evaluated at the time of the event. Prominent examples are key macroeconomic variables at the end of a fiscal year or quarter (see, e.g., Christodoulakis & Mamatzakis, 2009; Clements, 1997; Demetrescu et al., 2021), the wind speeds of hurricanes when they hit inhabited areas (Regnier, 2017), or the results of political elections (Kenett et al., 2018; Linzer, 2013; Selb & Munzert, 2016). The time periods ahead of the event are denoted as lead time. As time progresses, the fixed-event moves closer and the lead time decreases until the fixed-event occurs. The decreasing lead time and the "one-shot" character of fixed-event forecasting are key differences to conventional rolling event forecasting (Nordhaus, 1987). The generated forecasts and forecast errors are also different. In rolling event forecasting, each forecast refers to a different event while the lead time remains constant. In fixed-event forecasting, the forecasts refer to the same event and the lead time varies across the different forecasts.

Forecasting practice is often focused on evaluating the predictive performance of different models based on historic data and aims to identify the "best" technique with respect to a certain metric among a set of candidate approaches. Frequently, much time is spent by the analyst to select different methods and predictive algorithms, carefully train the methods, and craft an extensive framework for their evaluation. Examples are the M competitions

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(see, e.g., Makridakis et al., 1982, 1993; Makridakis & Hibon, 2000; Makridakis et al., 2018, 2022) or forecasting the vote shares obtained by different parties in political elections (Selb & Munzert, 2016). There are considerable pitfalls when evaluating forecasting systems based on the predictive performance only. In the worst case scenario, much time is spent in vain, when the chosen method does not produce proper forecasts (see, e.g., Dowding & Miller, 2019; Schrodt, 2014, for discussions of room for potential improvement in quantitative political analysis) for various reasons. First, considering historical data provides only a snapshot of the performance at a particular point in time. Second, pre-registering of forecasting studies is rare and the reported results may suffer from selective reporting where only favorable empirical evidence is included in the published version of a paper. Third, when judgment is involved in data pre-processing or forecasting, empirical evidence suggests that feedback about systematic bias is key to improving the forecasts (Petropoulos et al., 2017). Fourth, the chosen methods and predictive algorithms to set up a forecasting system rely on particular assumptions, which have certain implications for the obtained forecasts. Typically, discussing the assumptions receives only limited attention and empirically assessing the time series properties of forecasts is rare in practice. This highlights the importance of extending the scope of the analysis beyond the predictive performance to thoroughly investigating the properties of the underlying forecasting systems.

One such property is the efficiency of forecasting systems and builds on the rational expectations hypothesis (Muth, 1961). Among the assertions of the hypothesis is that, since information is scarce, all available information is employed each time a forecast is made. In efficient fixed-event forecasting systems, new information is incorporated instantly (Nordhaus, 1987). Efficiency also implies the absence of systematic bias (or reliability), revisions of the forecasts being martingale differences, and strict improvement of the forecast errors over time (Regnier, 2017). Depending on the system under consideration, at least some of these conditions are empirically testable based on a series of fixed-event forecasts (see, e.g., Chang et al., 2013; Clements, 1997; Clements et al., 2007; Clements & Taylor, 2001; Davies & Lahiri, 1995; Regnier, 2017).

A property related to the (in-)efficiency of a forecasting system is how information is adopted by the system (or the degree of information rigidity; Vereda et al., 2021). Theories on information rigidity postulate that data collection and processing is costly or that agents only observe noisy signals. In forecasting the results of political elections, election polls are frequently employed (Kenett et al., 2018; Petropoulos et al., 2022), which highlights judgment in forecasting as another source of uncertainty. The polls are carried out by different institutes and designed to depict the current political sentiment of voters towards parties. The polls are inherently judgmental: In order to ensure representativeness of the sample for the population of all eligible voters, (usually unpublished) sampling weights are employed by the institutes (Gelman, 2007; O'Muircheartaigh & Lynn, 1997). As such, election polls are in the spirit of the third role of judgment in forecasting outlined in Goodwin and Wright (1993): Information about the sample and its composition (used to weight the sample) and time series data (resulting representative election polls) are integrated to construct forecasts of party vote shares. Another form of judgment are the tendencies of different institutes towards publishing polls which are more favorable for certain parties (Shirani-Mehr et al., 2018). These forms of judgment affect the properties of poll-based forecasting systems and, to the best of our knowledge, a thorough analysis has not been carried out so far.

This paper addresses this gap and investigates the efficiency of election poll-based fixed-event forecasting systems for German federal state elections. We review fixed-event forecasting and the efficiency of forecasting systems and its implications. Empirical diagnostics and tests for least squares based martingale difference revisions and least absolute deviations based quantilegale difference revisions are provided. Both weak efficiency concepts are used to evaluate the election poll-based fixed-event forecasting systems for German state elections. Our results shed light on the question if the poll-based forecasting systems can be improved systematically and the related role of judgment in intentions surveys. The empirical evidence suggests that this is the case as weak efficiency of the forecasting systems is rejected. We also discuss alternative explanations for our findings beyond the rejection of weak efficiency.

Previous work on poll-based forecasting systems has quantified data quality (Meng, 2018), investigated construction-based biases and sampling errors (Mercer, 2016; Selb & Munzert, 2016; Tahk, 2015) and how to correct them with sampling weights (Gelman, 2007; Isakov et al., 2020; O'Muircheartaigh & Lynn, 1997) or different assumptions (Isakov et al., 2020), or considered the memory properties of election polls (Byers et al., 1997; Dolado et al., 2003). Other work has designed forecasting systems for particular types of elections (Aichholzer & Willmann, 2014; Linzer, 2013; Selb & Munzert, 2016) or analyzed the performance of poll-based forecasting for specific elections such as the 2016 US presidential election (Gelman & Azari, 2017), various regional elections in the US (Shirani-Mehr et al., 2018), and national elections in the US, UK, Canada, and Israel (Kenett et al., 2018).

The remainder of this paper is structured as follows. Section 2 introduces the relevant variables and processes for multi-period fixed-event forecasting systems. Section 3 details the properties and implications of least squares- and quantile-based weakly efficient forecasting systems. Section 4 summarizes empirical test equations for weak efficiency. Section 5 introduces the data and investigates the efficiency properties of poll-based fixed-event forecasting systems for German state elections. Section 6 summarizes the results and identifies strands for future research.

2. Multi-period fixed-event forecasting systems

In fixed-event forecasting, a target variable Y is forecast for a specific lead time λ before the fixed-event τ (Nordhaus, 1987), i.e., the target Y is a random variable

$$Y := Y_\tau \tag{1}$$

with realization $y \in [0, 1]$. A multi-period probability-forecasting system for the fixed-event Y (Regnier, 2017), for (equidistant or non-equidistant) lead times satisfying

$$\lambda_{\max} > \lambda_{\max} - 1 > \cdots > \lambda_1 > \lambda_0 = 0,$$

generates a sequence of forecast random variables

$$X_{\tau-\lambda} := X_{\tau|\tau-\lambda}, \quad \lambda_{\max} \geq \lambda \geq \lambda_1, \tag{2}$$

with realizations $(x_{\tau-\lambda_{\max}}, \ldots, x_{\tau-\lambda_1}) \in [0, 1]$. Here, $\lambda_{\max}$ denotes the maximum lead time and λ_1 denotes the minimum lead time. Thus, the final forecast is $X_{\tau-\lambda_1}$, before the fixed-event occurs at $\lambda_0 = 0$, and we set $X_{\tau-\lambda_0} = Y_\tau$. All information available at $\tau - \lambda$, when the fixed-event forecast is made, is contained in the information set $I_{\tau-\lambda}$, a sub-σ-field induced by the finite sequence of random variables $X_{\tau-\lambda_{\max}}, \ldots, X_{\tau-\lambda}$.

When new information is revealed, the information set and the fixed-event forecasts are updated (or revised; see Nordhaus, 1987). Recording the changes of (2) over time gives the process of fixed-event forecast revisions

$$Z_{\tau-\lambda} := Z_{\tau|\tau-\lambda}, \quad \lambda_{\max} - 1 \geq \lambda \geq \lambda_1, \tag{3}$$

with realizations $(z_{\tau-\lambda_{\max}+1}, \ldots, z_{\tau-\lambda_1}) \in [-1, 1]$, and, for $\lambda_s < \lambda_t$, $Z_{\tau-\lambda_s} = X_{\tau-\lambda_s} - X_{\tau-\lambda_t}$ denotes the change of the forecast between two consecutive lead times. All information available at $\tau - \lambda$ is contained in the information set $I^*_{\tau-\lambda}$, a sub-σ-field induced by the finite sequence of random variables $Z_{\tau-\lambda_{\max}+1}, \ldots, Z_{\tau-\lambda}$. Due to $X_{\tau-\lambda_0} = Y_\tau$, the final forecast revision equals

$$Z_{\tau-\lambda_0} = X_{\tau-\lambda_0} - X_{\tau-\lambda_1} = Y_\tau - X_{\tau-\lambda_1}. \tag{4}$$

The error of a forecast made λ periods ahead of the fixed-event is defined as

$$U_{\tau-\lambda} := U_{\tau|\tau-\lambda} = Y_\tau - X_{\tau-\lambda}, \quad \lambda_{\max} \geq \lambda \geq \lambda_1, \tag{5}$$

implying that the final forecast error is zero, $U_{\tau-\lambda_0} = Y_\tau - X_{\tau-\lambda_0} = 0$. For lead time λ_1, (4) implies $U_{\tau-\lambda_1} = Z_{\tau-\lambda_0}$. Iterating this computation until lead time λ, we obtain

$$U_{\tau-\lambda} = Z_{\tau-\lambda_0} + Z_{\tau-\lambda_1} + \cdots + Z_{\tau-\lambda+1}. \tag{6}$$

Thus, $U_{\tau-\lambda}$ is equal to the sum of all forecast revisions made after the forecast. Equivalently, we can decompose the fixed event into the initial forecast and the cumulative revisions,

$$\begin{aligned} X_{\tau-\lambda_0} &= X_{\tau-\lambda} + Z_{\tau-\lambda_0} + Z_{\tau-\lambda_1} + \cdots + Z_{\tau-\lambda+1} \\ &= X_{\tau-\lambda_{\max}} + Z_{\tau-\lambda_0} + Z_{\tau-\lambda_1} \\ &\quad + \cdots + Z_{\tau-\lambda+1} + \cdots + Z_{\tau-\lambda_{\max}+1}. \end{aligned} \tag{7}$$

Example. For the election polls considered in this paper, the target $Y_{\tau,r(p)}$ is the election outcome for party p in region r at election time τ. A particular election constitutes a forecasting system, (r, τ). For each party p in (r, τ), we can examine the properties of the forecast random variables $X_{\tau-\lambda}$ and the corresponding forecast revisions $Z_{\tau-\lambda}$. For example, we consider a forecasting system (r = Bavaria, τ = 2018) for the six dominating parties competing for the seats of the Bavarian senate. For ease of notation we omit p and r whenever the context is clear.

3. Weakly efficient forecasting systems

For our presentation of the theory of weakly efficient fixed-event multi-period probability-forecasting systems, we introduce some standard econometric notation: Let $X_{\tau-\lambda}$ be defined on a complete probability space $(\Omega, \mathcal{F}, \mathbb{P})$ and let $\mathcal{F}_{\tau-\lambda}$ be the filtration to which $X_{\tau-\lambda}$ is adapted, where $\mathcal{F}_{\tau-\lambda}$ is the sub-σ-field generated by $I_{\tau-\lambda}$.

Gneiting and Raftery (2007) considered the theoretical properties of a forecasting system based on the minimization of a proper scoring rule. In forecasting practice, a scoring rule $L(\cdot)$ is tailored to the forecasting task (Gneiting, 2011a) and scores resulting from evaluating $L(\cdot)$ based on given data are compared directly across candidate methods. Alternatively, hypotheses about their relative predictive performance are tested (Diebold, 2015; Diebold & Mariano, 2002). This yields a snapshot of the predictive performance of a forecasting system at a particular time period, while in theory, a forecasting system can be used to create an infinite set of point forecasts for different events (Regnier, 2017).

The fixed-event forecast $X_{\tau-\lambda}$ minimizing expected loss (risk) $\mathbb{E}[L(Y_\tau, X_{\tau-\lambda}) \mid I_{\tau-\lambda}]$, given all available information up to time period $\tau - \lambda$, leads to a minimal condition for weak efficiency of a forecasting system $\mathbb{E}[L(Y_\tau, X_{\tau-\lambda}) \mid I_{\tau-\lambda}] \leq \mathbb{E}[L(Y_\tau, g(X_{\tau-\lambda})) \mid I_{\tau-\lambda}]$, for all $\mathcal{F}_{\tau-\lambda}$-measurable weighting functions $g(\cdot)$. This condition requires that the current forecast cannot be improved systematically if the entire history of past forecasts generating the information $I_{\tau-\lambda}$ is used, compared to using only the most recent forecast $X_{\tau-\lambda}$.

Example (ctd.). In our empirical example of the forecasting system (r, τ), the event "polls for election at time τ for party p in region r" has a specified starting date at $\tau - \lambda_{\max}$. Obviously, in this case the initial filtration $\mathcal{F}_{\tau-\lambda_{\max}}$ is not the empty set as typically previous information concerning $p(r)$ is available at poll time $\tau - \lambda_{\max}$. As Nordhaus (1987) pointed out the practical limitations of including this information by using an extended information set $J_{\tau-\lambda}$, where $I_{\tau-\lambda} \subset J_{\tau-\lambda}$, we will not pursue the underlying concept of strong efficiency. To simplify our notation, we will not distinguish between $I_{\tau-\lambda}$ and $I^*_{\tau-\lambda}$ in the following, although both include only the finite past and may differ.

Depending on the choice of a particular scoring rule, orthogonality conditions can be derived and analyzed using statistical tests. We start with least squares loss, which is common in the literature. We then discuss whether asymmetrically weighted least absolute deviations (or quantile) loss is a viable alternative in this context.

3.1. Least squares loss fixed-event forecasting

All studies known to the authors consider L_2-loss for fixed-event forecasting. Assuming existence of all expectations in this section, expected loss $\mathbb{E}[L_2(Y_\tau, X_{\tau-\lambda}) \mid I_{\tau-\lambda}] = \mathbb{E}[|Y_\tau - X_{\tau-\lambda}|^2 \mid I_{\tau-\lambda}]$ is well known to be minimal, if $X_{\tau-\lambda} = \mathbb{E}[Y_\tau \mid I_{\tau-\lambda}]$, or

$$\mathbb{E}[U_{\tau-\lambda} \mid I_{\tau-\lambda}] = 0 \tag{8}$$

holds almost surely with respect to the probability measure $\mathbb{P}$ ($\mathbb{P}$-*a.s.*). This is equivalent to the weak efficiency

of fixed-event forecasting systems as introduced by Nordhaus (1987) and condition (8) implies that forecasts are dynamically complete (linear) L_2 projections.

As a consequence, for lead times satisfying $\lambda_s < \lambda_t$, the population orthogonality condition

$$\mathbb{E}[Z_{\tau-\lambda_s} g(I_{\tau-\lambda_t})] = 0, \tag{9}$$

holds if and only if forecasts in a weakly efficient multi-period probability-forecasting system are martingales (e.g., Nordhaus, 1987; Regnier, 2017). Considering (7) and (8), this is easy to verify as weakly efficient forecasting systems generate forecasts that are random walks with drift (i.e., the initial forecast $X_{\tau-\lambda_{\max}}$) and mean independent increments (i.e., the forecast revisions) (e.g., Clements, 1997; Escanciano & Lobato, 2009). This implies, for $\lambda_s < \lambda_t$, the martingale null hypothesis

$$\mathbb{E}[X_{\tau-\lambda_s} \mid I_{\tau-\lambda_t}] = X_{\tau-\lambda_t} \ \ \mathbb{P}\text{-}a.s. \tag{10}$$

In analogy to test equations for forecast rationality (see, e.g., Mincer & Zarnowitz, 1969; Mishkin, 1981; Zarnowitz, 1985), some of the test equations for weak efficiency of fixed-event forecasting systems via the martingale difference null hypothesis contain an intercept (see, e.g., Clements, 1997, equation (4), Clements et al., 2007, equation (18), Chang et al., 2013, equation (1)). A theoretical justification can be seen in the test proposed by Phillips and Jin (2014), who argue that in many economic applications, time series X_t may contain a small (but possibly negligible) drift μt, where $\mu \in \mathbb{R}$ is an evaporating intercept $\mu = \mu_0 n^{-\gamma}$ depending on the sample size n and the parameter γ. If $\gamma > 0.5$, the drift does not dominate the stochastic trend and we can consider a martingale hypothesis $X_t = \mu + \theta X_{t-1} + u_t$ for $\theta = 1$ and obtain $X_t = X_0 + \mu t + \sum_t u_t$ (see Phillips & Jin, 2014, equations (5)-(7)). An alternative null hypothesis therefore allows the forecasts $X_{\tau-\lambda}$ to have a non-zero and unknown unconditional expected value and is given by $\mathbb{E}[X_{\tau-\lambda_s} - \mu \mid I_{\tau-\lambda_t}] = X_{\tau-\lambda_t}$ $\mathbb{P}$-*a.s.*, if and only if $\mathbb{E}[(Z_{\tau-\lambda_s} - \mu) g(I_{\tau-\lambda_t})] = 0$.

In fixed-event forecasting, most studies (e.g., Clements, 1997; Clements & Taylor, 2001; Nordhaus, 1987; Regnier, 2017) conduct their empirical analyses based on forecast revisions $Z_{\tau-\lambda}$ and therefore use a martingale difference null hypothesis, for $\lambda_s < \lambda_t$,

$$\mathbb{E}[Z_{\tau-\lambda_s} - \mu \mid I_{\tau-\lambda_t}] = 0 \ \ \mathbb{P}\text{-}a.s. \tag{11}$$

While Escanciano and Lobato (2009) examined a variety of tests of the martingale and martingale difference hypotheses and their asymptotic size and power, as well as their robustness to alternatives, the literature on fixed-event forecasts generally has to rely on very small sample sizes, since the number of expert surveys or election polls for each event under consideration is typically limited. A review of empirical results and a thorough discussion of the test equations to investigate the martingale and martingale difference hypotheses can be found in Chang et al. (2013). When empirically evaluating (fixed-event) forecasting systems, the entire literature known to the authors assumes that $g(\cdot)$ in (9) is linear and univariate, e.g., $g(I_{\tau-\lambda}) = Z_{\tau-\lambda-j}$ for some $j \geq 1$. Following this literature investigating linear measures of dependence for finite-dimensional information sets (see Escanciano & Lobato, 2009, Sec. 3.1), we replace (9) by

$$\mathbb{C}\text{ov}[Z_{\tau-\lambda}, Z_{\tau-\lambda-j}] = \mathbb{E}[(Z_{\tau-\lambda} - \mu) Z_{\tau-\lambda-j}] = 0, \quad j \geq 1. \tag{12}$$

A correlogram can be computed from $\mathbb{C}\text{ov}[Z_{\tau-\lambda}, Z_{\tau-\lambda-j}]$ to study Box tests, or Dickey–Fuller tests can be used to investigate whether forecasts are non-stationary. If forecast revisions are martingale differences, condition (12) holds, and we can call such a forecasting system L_2-weakly efficient.

An investigation of the efficiency of forecasting systems should consider alternative choices for proper scoring rules if the loss function of the forecaster is unknown (Patton & Timmermann, 2007b). In particular, the empirical evidence cited in our paper as well as our own empirical analyses of forecasting systems generated by German state election polls indicate that the L_2-weak efficiency in the sense of Nordhaus does not seem to reflect the underlying loss function of the forecasters correctly on a regular basis. As discussed by Clements (1997), forecaster's loss may depend asymmetrically on positive and negative forecast errors. A simple generalization would be to consider expectiles (see, e.g., Kneib, 2013) to investigate asymmetric L_2-weak efficiency. Due to the simple interpretation of quantiles (compared to expectiles), we will use the L_α-weak efficiency approach based on α-quantile-loss in the following.

3.2. Quantile loss fixed-event forecasting

Even though there is no widely recognized L_1- or L_α-based equivalent to the martingale hypothesis, the application of median or quantile regression has its place in time series analysis, especially in the treatment of non-normally distributed processes in finance, macroeconomics, risk assessment (see, e.g., Engle & Manganelli, 2004), or in modeling risk attitudes (Bassett et al., 2002). Linton and Whang (2007) discuss economic and statistical reasons for using quantile loss to study (directional) predictability and emphasize its natural appeal for this task (in finance and beyond). In our problem at hand, the concept of a mediangale as defined below may be related to a martingale, if, for example, forecasts are (conditionally) symmetric. In general, however, while a forecasting system may be weakly efficient in L_2 sense, it may be inefficient in L_1 (or L_α) sense, or vice versa.

While the martingale concept requires integrability of forecasts, this may not be a viable assumption in some applications (see, e.g., Linton & Whang, 2007, p. 253f). On the other hand, for a given α ($0 < \alpha < 1$), we can define a conditional α-quantile function Q_α of $X_{\tau-\lambda}$ with respect to a sub-σ-field $\mathcal{F}_{\tau-\lambda}$, as a random variable such that $\mathbb{P}(X_{\tau-\lambda} - Q_\alpha \leq 0) \geq \alpha$, $\mathbb{P}$-*a.s.* Since every ($\mathcal{F}_{\tau-\lambda}$-measurable) random variable has a conditional quantile function, we can use the usual notation $\mathbb{Q}_\alpha[\cdot|\cdot]$. The expected L_α-loss is given by

$$\begin{aligned}&\mathbb{E}[L_\alpha(Y_\tau, X_{\tau-\lambda}) \mid I_{\tau-\lambda}] \\ &\quad = \mathbb{E}[(Y_\tau - X_{\tau-\lambda}) \cdot \psi_\alpha(Y_\tau - X_{\tau-\lambda}) \mid I_{\tau-\lambda}],\end{aligned} \tag{13}$$

where $\psi_\alpha(Y_\tau - X_{\tau-\lambda}) := \alpha - 1(Y_\tau - X_{\tau-\lambda} < 0)$ and $1(A)$ is the usual indicator function of the set A. Expected quantile loss (13) is well known to be minimal, if the forecast is correctly specified (i.e., dynamically complete) in the sense that $X_{\tau-\lambda} = \mathbb{Q}_\alpha[Y_\tau \mid I_{\tau-\lambda}]$ holds $\mathbb{P}$-*a.s.*. Then,

$$\mathbb{Q}_\alpha[U_{\tau-\lambda,\alpha} \mid I_{\tau-\lambda}] = 0 \ \ \mathbb{P}\text{-}a.s., \tag{14}$$

where we define (Koenker & Xiao, 2004) $U_{\tau-\lambda,\alpha} := Y_\tau - \mathbb{Q}_\alpha[Y_\tau \mid I_{\tau-\lambda}]$, implying that forecasts are dynamically complete (nonlinear) L_α projections. An insightful discussion of the conditional quantile function, the law of iterated quantiles and the nature of L_α projections is provided in de Castro et al. (2023). While it seems difficult to derive a theory analogous to Nordhaus' weak efficiency concepts of forecasts with respect to L_2-losses, there are strongly related robust (quantile-based) alternatives to autocovariance-based concepts such as (12).

The random variable $\psi_\alpha(V_t)$, sometimes denoted as quantile score or quantile indicator, is well known in quantile time series analysis: It captures the structure of dependence and heterogeneity of a random variable V_t under L_α-loss and all of its moments exist under standard conditions for quantile estimation. As a consequence, its variances and auto-covariances can be easily calculated under general conditions (see, e.g., Oberhofer & Haupt, 2005, p. 247). Assuming stationary $Z_{\tau-\lambda}$ and $\mu_\alpha := \mathbb{Q}_\alpha[Z_{\tau-\lambda}]$, Linton and Whang (2007) define the quantilogram

$$\frac{\mathbb{E}[\psi_\alpha(Z_{\tau-\lambda} - \mu_\alpha)\,\psi_\alpha\left(Z_{\tau-\lambda-j} - \mu_\alpha\right)]}{\mathbb{E}[\psi_\alpha^2(Z_{\tau-\lambda})]} \tag{15}$$

and use it to investigate the null hypothesis, for $\lambda_s < \lambda_t$,

$$\mathbb{E}[\psi_\alpha\left(Z_{\tau-\lambda_s} - \mu_\alpha\right) \mid I_{\tau-\lambda_t}] = 0 \ \ \mathbb{P}\text{-}a.s. \tag{16}$$

Note that this hypothesis tests whether revisions are α-quantilegale differences, or, equivalently, as noted by Patton and Timmermann (2007b, p. 1176), that $\psi_\alpha(Z_{\tau-\lambda} - \mu_\alpha)$ are martingale differences. The definition of the sign-based random variable $\psi_\alpha(\cdot)$ indicates its usefulness in capturing the nature of a "jagged efficient forecast" (as Nordhaus, 1987, p. 668, puts it to explain the nature of martingales). However, non-rejection of (16) does not mean that the revisions $Z_{\tau-\lambda} - \mu_\alpha$ are martingale differences. If forecast revisions are α-quantilegale differences, condition

$$\mathbb{E}[\psi_\alpha(Z_{\tau-\lambda} - \mu_\alpha)\,\psi_\alpha\left(Z_{\tau-\lambda-j} - \mu_\alpha\right)] = 0, \quad j \geq 1, \tag{17}$$

holds, and we call such a forecasting system L_α-weakly efficient. Thus, the quantilogram can be computed to study Linton-Whang variants of Box tests (see Linton & Whang, 2007, Sec. 3). While Dickey–Fuller tests are frequently used for investigating the martingale hypothesis, a KPSS test for stationarity based on the quantile indicator proposed by Busetti and Harvey (2010) may be used in a similar spirit to investigate the α-quantilegale difference hypothesis.

4. Empirical checks of weak efficiency

Least squares loss. The literature assuming (12) employs first-order autoregressive test equations for forecast revisions which are estimated by OLS (Nordhaus, 1987). For poll-based forecasting systems designed to forecast the vote shares of parties $p = 1, \ldots, P$, the simple dynamic panel equation (see Pesaran, 2015, p. 724) with one-way (party) error component and heterogenous slope coefficient (employing the usual time series notation)

$$Z_{p,t} = \theta_p Z_{p,t-1} + \mu_p + \epsilon_{p,t}, \tag{18}$$

is a more efficient alternative to estimating separate test equations for each party by OLS (see Clements, 1997, equation (4), p. 227). The θ_p are party-specific slope coefficients that reflect varying dynamic adjustment across the parties and the composite error term $u_{p,t} = \mu_p + \epsilon_{p,t}$ can be decomposed into a party-specific intercept μ_p (modeled via party-dummies) and an idiosyncratic remainder term $\epsilon_{p,t}$. Under the null that the forecasting system is weakly efficient, $\theta_p = 0, \forall p$, and the generated forecasts follow a random walk with (possibly negligible party-specific) deterministic drift μ_p (see Phillips & Jin, 2014, p. 539).

Two special cases of test Eq. (18) are commonly employed in the literature. Assuming that the slope coefficients of all parties are identical (see, e.g., Baltagi, 2021, Sec. 8.1) results in the dynamic panel equation with homogeneous slope coefficient

$$Z_{p,t} = \theta Z_{p,t-1} + \mu_p + \epsilon_{p,t}. \tag{19}$$

For a sufficiently large number of longitudinal observations $t = 1, \ldots, T$, the slope coefficient of (19) can be estimated consistently by OLS when incorporating party-specific dummies (Kiviet, 1995). Imposing homogeneous intercepts, the pooled OLS estimator

$$Z_{p,t} = \theta Z_{p,t-1} + \mu + \epsilon_{p,t} \tag{20}$$

is obtained (see Clements, 1997, p. 232f). A generalization of test Eq. (18) is given by the dynamic two-way error component model

$$Z_{p,t} = \theta_p Z_{p,t-1} + \mu_p + c_t + \epsilon_{p,t}, \tag{21}$$

which employs the error decomposition $u_{p,t} = \mu_p + c_t + \epsilon_{p,t}$ into party-specific intercept μ_p, time-specific intercept c_t (modeled via time dummies), and the idiosyncratic remainder term. The time-specific intercept, for example, can absorb the effects of different weighting schemes used for different surveys or employed by different polling institutes (Gelman, 2007; O'Muircheartaigh & Lynn, 1997) or potential measurement errors occurring over the lead times (see, e.g., Patton & Timmermann, 2011). Based on the test Eqs. (18) to (21), we compute OLS estimators (see Nordhaus, 1987, Sec. B) and discuss their respective significance separately for every forecasting system (see Regnier, 2017, Sec. 3.2).

Example (ctd.). The slope coefficients and intercepts of the test Eqs. (18), (19), and (20) are illustrated in Fig. 1 for the forecasting system (Bavaria (BY), 2018). The upper left display shows the observations: Note that since the polling institutes state the original data in percentage points (at most in one digit precision), the observations are jittered in order to create this plot (i.e., a small amount of noise is added to the abscissa and ordinate values).

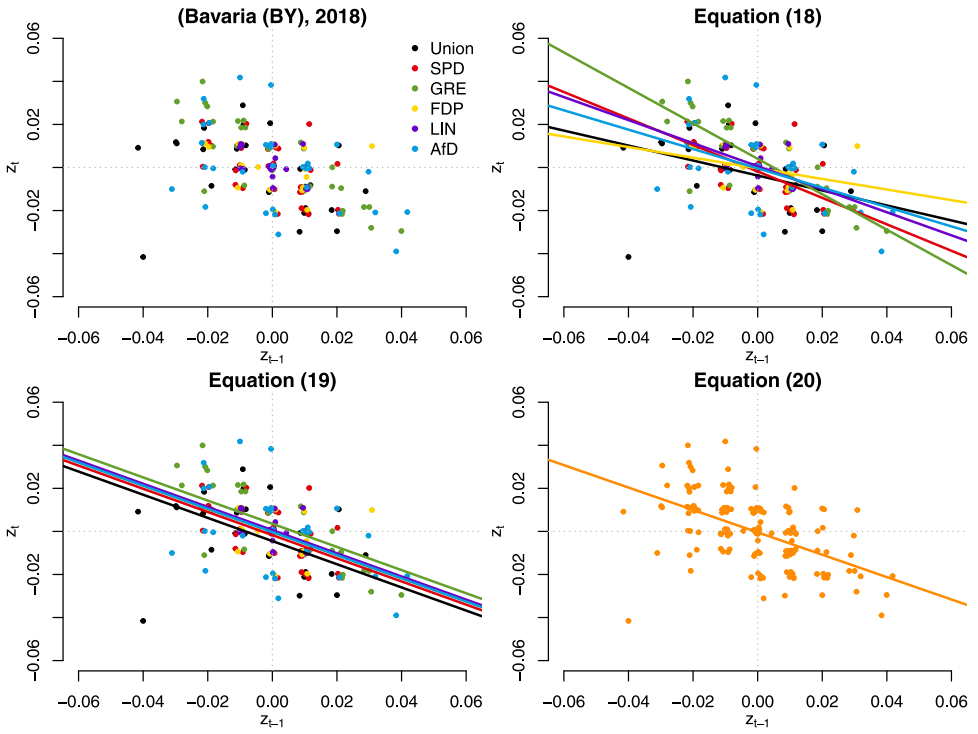


Fig. 1. Scatterplot of revisions Z_t versus lagged revisions Z_{t-1} and OLS estimates for forecasting system (Bavaria (BY), 2018) displayed separately for test Eqs. (18), (19), and (20).

The upper right display shows the estimates based on the test equation with party-specific intercept and slope. In the lower displays the slope coefficient is homogeneous across parties. The parallel lines in the lower left display are shifted by the party-specific intercepts, while the lower right display illustrates results when pooling over parties (i.e., homogeneous intercept and slope across all parties). In comparison to the fitted lines shown in the upper right display, the estimation of Eq. (21) would additionally incorporate (lead) time-specific dummies, shifting the regression lines for each party up/down (by the respective time-specific intercept).

Quantile loss. Our empirical results in Section 5 below reflect existing findings in the literature that cast doubt on the L_2-weak efficiency of the forecasting systems investigated. The small sample sizes are particularly problematic given that estimators and tests for investigating martingales and martingale differences heavily rely on asymptotic approximations. In addition, this problem renders the potential relevance of higher-order dynamics even more problematic, as the proposed concepts rely on correctly specified orthogonality conditions. Further, nonlinear forms of dependence may be a problem on top of all others. As this is all well known, we try to pursue an alternative to the existing least squares based approaches using the pinball loss function of quantile estimation (see Koenker, 2017, p.156 and Gneiting, 2011b, p. 199–200). Koenker and Xiao (2004, 2006) detail the quantile estimation of test Eq. (18) under the quantile restriction of dynamic completeness (14).

5. Empirical illustration: election forecasting systems in Germany

This section introduces the German federal state election poll data and details the evaluation of three exemplary poll-based multi-period probability forecasting systems for weak efficiency. Results for further forecasting systems are provided in Appendix.

5.1. Multi-period state election probability forecasting systems

We consider election polls where respondents indicate who they would vote for if the elections were held next Sunday. The polls were collected by the different pollsters (opinion research institutes/companies) summarized in Table 1, with the four largest providers accounting for roughly 72% of the polls. By far the most common type of poll in the dataset were telephone polls (64.4%) followed by telephone-online-mix polls (8.7%) and online polls (7.2%), while face-to-face and telephone-short-message polls rarely occured (<1%). For 233 polls (18.9%), the type was not available. In the German national electoral system, voters can cast two votes. The so-called "Erststimme" and the "Zweitstimme". With the former, a candiate is elected into parliament directly, while the latter is given to a party and determines the relative number of seats a party wins in parliament. This electoral system is also employed in German state elections – with the exception of Baden-Wuerttemberg and Saarland, where voters can cast only one vote and Bremen and Hamburg,

Table 1
Acronyms, pollsters (institution/company name), number of provided election polls, and average poll participants. Pollsters providing less than 20 polls are included in category Other.

Acronym	Opinion research institution/company	Number of polls	Average participants
FGW	Forschungsgruppe Wahlen e.V.	151	1144
Forsa	Forsa GmbH	199	1081
GMS	GMS Dr. Jung GmbH	53	1032
IfM	Leipziger Institut für Marktforschung	26	899
Infratest	Infratest dimap GmbH	394	1046
INSA	INSA-Consulere GmbH	145	1031
Psephos	Institut für Wahlforschung und Sozialwissenschaft GmbH	30	975
Verian	Kantar GmbH	119	930
YouGov	YouGov Deutschland GmbH	22	1028
Other	Other	94	867

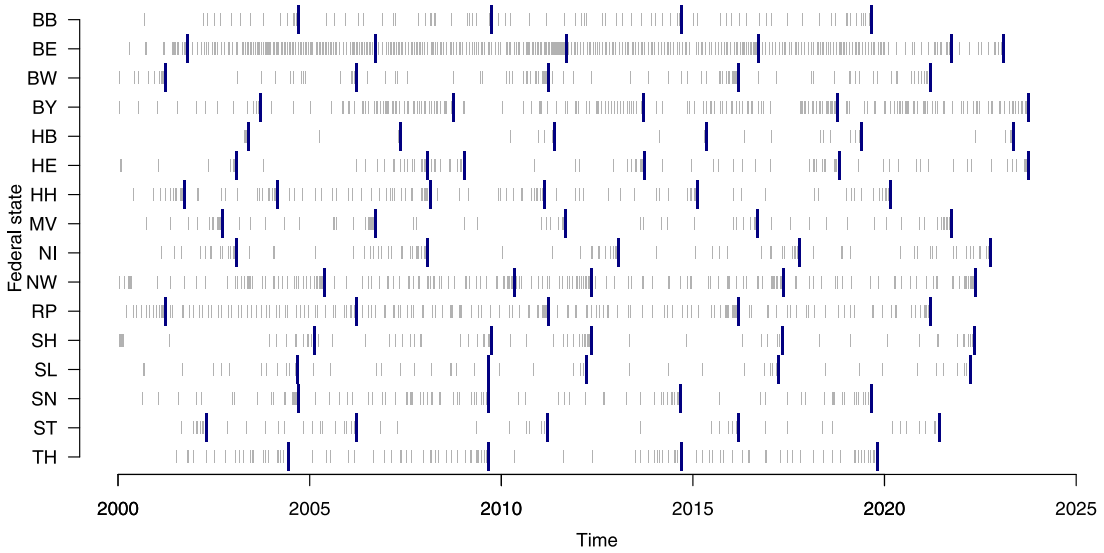


Fig. 2. Poll dates (ticks) and election dates (lines) for all 16 German federal states.

where voters can currently cast 5 and 10 votes (see https://www.wahlrecht.de/landtage/ Cantow et al., 2021). The polls target the party-preference (the "Zweitstimme" in most states) and are conducted over several days, where we use the date of publication as the date of each election poll. The majority (75%) of polls are completed after 7 days or less and subsequently published within 4 days, with minor differences between the different types of polls. To obtain the election poll data for each poll date, the voting intentions of the survey participants are then aggregated for each party.

The database is publicly available at www.wahlrecht.de (Cantow et al., 2021). We collected polling data for all German federal state elections from January 2000 to February 2024. Despite the availability of the poll and corresponding meta-data, the properties of German federal state election polls were still not studied extensively. The multiparty system and the changing party landscape over the last two decades, with the rise of the formerly small party the Greens (Grüne), the foundation of the AfD and the decline of the peoples's parties CDU/CSU and SPD, provide further intrigue. The stream of polls for every German state is depicted by vertical ticks in Fig. 2, vertical lines indicate state elections.

The examined multi-period probability forecasting systems (r, τ) are designed to forecast the vote shares of parties in a given state r at time τ. We restrict our analysis to the major parties that frequently participate in state elections in all states and that regularly exceed the five percent threshold. This hurdle is mandatory for a party to enter parliament, with the exception of parties representing ethnic minorities. The state abbreviations indicated on the ordinate are listed in Table 2. Table 3 lists the parties, their acronyms, and traditional colors used to represent all party-specific information in the following visualizations. The table also includes information on the number of times the parties participated in the 81 state elections covered by the dataset and the number of times the parties exceeded the 5% threshold.

5.2. Three state election forecasting systems

As leading examples, we consider the three poll-based multi-period fixed-event probability forecasting systems $(r, \tau) = (\text{BY}, 2018)$, $(\text{BE}, 2011)$, and $(\text{NW}, 2005)$ and evaluate their L_2- and L_α-weak efficiency, respectively. We selected the leading examples such that they covered enough observations to expect valid results. The city state

Table 2
Abbreviations of German federal states.

Abbr.	Federal state	Abbr.	Federal state
BB	Brandenburg	NI	Lower Saxony
BE	Berlin	NW	Northrhine-Westphalia
BW	Baden-Wuerttemberg	RP	Rhineland Palatinate
BY	Bavaria	SH	Schleswig-Holstein
HB	Hanseatic City of Bremen	SL	Saarland
HE	Hesse	SN	Saxony
HH	Hanseatic City of Hamburg	ST	Saxony-Anhalt
MV	Mecklenburg Western Pomerania	TH	Thuringia

Table 3
Acronyms, party names, traditional party colors, elections the parties participated in, and number of times 5% threshold was exceeded.

Acronym	Party	Color	Elections participated	Threshold exceeded
Union	Christian Democratic/Social Union (CDU/CSU)	Black	81	81
SPD	Social Democratic Party (SPD)	Red	81	81
GRE	Federation 90/The Greens (GRÜNE)	Green	81	71
FDP	Free Democratic Party (FDP)	Yellow	81	54
LIN	The Left Party (Die LINKE)	Purple	81	47
AfD	Alternative for Germany (AfD)	Light blue	34	32

Table 4
Maximum lead times (in days) and corresponding number of polls for forecasting systems (BY, 2018), (BE, 2011), and (NW, 2005).

State	Election date	$\lambda_{\max} = 720$	$\lambda_{\max} = 365$	$\lambda_{\max} = 180$	$\lambda_{\max} = 90$
BY	2018–10–14	33	31	21	13
BE	2011–09–18	53	37	26	15
NW	2005–05–22	41	32	26	21

Table 5
Federal states r, election dates τ, and corresponding vote shares $y_{\tau,r(p)}$ per party (in %), as well as voter turnout (VT) for forecasting systems (BY, 2018), (BE, 2011), and (NW, 2005).

State	Election date	Union	SPD	GRE	FDP	LIN	AfD	VT
BY	2018–10–14	37.2	9.7	17.5	5.1	3.2	10.2	72.4
BE	2011–09–18	23.4	28.3	17.6	1.8	11.7		60.2
NW	2005–05–22	44.8	37.1	6.2	6.2	0.9		63.0

BE was selected, since three of the 16 federal states are city states. Due to the rise of the AfD in recent years, we also wanted to show results before and after the party was founded.

We consider only polls published within the last $\lambda_{\max} = 365$ days ahead of each state election to reflect the structure of the data and to cover actual campaign effects and voter sentiment. The number of available polls is given in Table 4 for $\lambda_{\max}$ denoted as alternative lead times. The table illustrates the trade-off between recency of the polls and sample size for evaluating the forecasting systems. When extending the lead time from $\lambda_{\max} = 365$ to $\lambda_{\max} = 720$, for example, two (BY), 16 (BE), and nine (NW) additional polls are available. Similar patterns hold for the other federal states and the number of polls generally increases as the election date approaches.

Table 5 shows the vote shares (i.e., realized target values $y_{\tau,r(p)}$) of the parties in the state elections that the three forecasting systems are targeting together with the election date and voter turnout (VT). Note that the party AfD was founded in 2013 and did not participate in elections held in previous years. For the party Die Linke, which was founded in 2007 when two left parties were joined together, the earlier results and polls correspond to either one (or all) of the founding parties. In the three selected elections, the parties Union, SPD, and GRE obtained roughly 2/3 or more of the votes that were cast in the elections.

Table 6 shows descriptive statistics that characterize the forecast revisions obtained from the three forecasting systems. For all three systems, the central tendency of the empirical distribution is close to zero and there are only minor differences between the mean and median for some parties. The standard deviations of the forecast revisions generally exceed the means and medians for the different parties.

Fig. 3 (top display) shows the realized forecasts x_t for each of the six parties generated by the forecasting system (BY, 2018). The other displays in Fig. 3 show the corresponding realized forecast revisions z_t separately for all six parties. As the forecast revisions result from first differencing the forecasts, the number of forecasts always exceeds the number of forecast revisions by one observation. Despite the similarity of Fig. 3 to Figure 7 in Regnier (2017) for one US election, there is a key

Table 6
Descriptives of forecast revisions for forecasting systems (BY, 2018), (BE, 2011), and (NW, 2005).

State	Year	Statistic	Union	SPD	GRE	FDP	LIN	AfD
BY	2018	Minimum	−0.0400	−0.0200	−0.0300	−0.0200	−0.0100	−0.0400
		Mean	−0.0035	−0.0006	0.0019	0.0005	0.0003	0.0000
		Median	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
		Maximum	0.0300	0.0200	0.0400	0.0300	0.0100	0.0400
		Std. dev.	0.0182	0.0134	0.0199	0.0107	0.0067	0.0198
		Skewness	−0.3055	−0.0502	0.0893	0.2483	−0.0343	0.2346
		Kurtosis	2.3070	1.8345	1.8025	3.3932	2.2356	2.5870
BE	2011	Minimum	−0.0500	−0.0600	−0.0400	−0.0200	−0.0400	
		Mean	0.0008	0.0008	−0.0019	−0.0005	−0.0008	
		Median	0.0000	0.0000	−0.0100	0.0000	0.0000	
		Maximum	0.0500	0.0500	0.0500	0.0200	0.0400	
		Std. dev.	0.0252	0.0248	0.0233	0.0087	0.0192	
		Skewness	0.0568	−0.0607	0.9411	0.1993	0.0238	
		Kurtosis	2.1338	2.7489	3.2222	2.6954	2.4745	
NW	2005	Minimum	−0.0300	−0.0500	−0.0300	−0.0100		
		Mean	−0.0012	0.0006	−0.0003	0.0000		
		Median	0.0000	0.0000	0.0000	0.0000		
		Maximum	0.0300	0.0600	0.0200	0.0200		
		Std. dev.	0.0134	0.0221	0.0143	0.0073		
		Skewness	0.0667	0.6997	−0.2299	0.5064		
		Kurtosis	3.7943	4.1378	2.4084	3.3058		

difference in the underlying methodology: While the latter employ daily (i.e., equidistant) forecasts (and forecast revisions) published by six major news outlets with many zero-revisions, our forecasts and forecasts revisions are non-equidistant and there is only a small share of zero-revisions.

Analogous to Fig. 3, the forecasts and forecast revisions are visualized for the two forecasting systems (BE, 2011) and (NW, 2005) in Figs. 4 and 5. For (BE, 2011), realized forecasts and realized forecast revisions were generated for five parties. For (NW, 2005), realized forecasts and realized forecast revisions were generated for four parties. The party LIN was omitted from Fig. 5 as it was assigned many zeros by the forecasting system (NW, 2005) – preceding a vote share close to zero in the election. The following stylized facts when comparing Figs. 3, 4, and 5 are worth mentioning: The most pronounced changes in realized forecasts over lead times are observable for (BY, 2018) for the parties Union (from approx. 0.45 to 0.37) and GRE (from approx. 0.12 to 0.2) – resulting in the sum of forecast revisions over the considered lead time being negative for the former party and positive for the latter. Overall, the most pronounced realized forecast revisions across the parties occur for (BE, 2011) followed by (BY, 2018) and (NW, 2005).

5.3. Weak efficiency of the three state election forecasting systems

We employ the realized forecast revisions to empirically evaluate the efficiency of the three poll-based fixed-event forecasting systems (BY, 2018), (BE, 2011), and (NW, 2005). Table 7 reports the slope coefficient OLS estimates and corresponding standard errors for the test equations with party-specific intercept and slope coefficient (18), homogeneous slope coefficient and party-specific intercept (19), and when pooling across parties (20). We omit the coefficient estimates for test Eq. (21) from the table, as the results and conclusions are similar to the displayed ones. Summarizing the findings for the three forecasting systems in Table 7, two main empirical insights emerge.

First, all coefficient estimates in the table are negative for all forecasting systems – with one exception for the party-specific coefficient of Union in (NW, 2005). The negative slope estimate is obtained for the forecasting systems for all German state elections since 2000 when test Eq. (19) is used and for all but three forecasting systems, when test Eq. (20) is used (see Appendix, Tables A.8, A.9). While Clements and Taylor (2001) and Nordhaus (1987) propose interpretations for the sign of the slope coefficient estimate, which relate to "forecast smoothing", Chang et al. (2013) illustrates that additional information is required to infer any conclusions from the sign. Further analysis in this direction might be insightful, when incorporating additional information on the quality of the data (Meng, 2018), such as the representativeness of the sample for the population of all eligible voters (Gelman & Azari, 2017), whether the poll values published by institutes systematically favor certain parties over others (Shirani-Mehr et al., 2018), or how the raw data were processed by weighting schemes to produce the reported poll values (Gelman, 2007; O'Muircheartaigh & Lynn, 1997).

Second, the empirical evidence for the forecasting systems discussed in Table 7 casts doubt on L_2-weak efficiency for the three considered forecasting systems: The majority of the reported coefficient estimates and standard errors indicate a rejection of the null hypothesis that past revisions and current revisions are orthogonal. This is clearly the case with the two test Eqs. (19) and (20) for all three elections – even if the results for test Eq. (18) show that there can be party- and forecasting system-specific differences. Similar interpretations and conclusions result from the evaluation of the forecasting systems for all German state elections from January

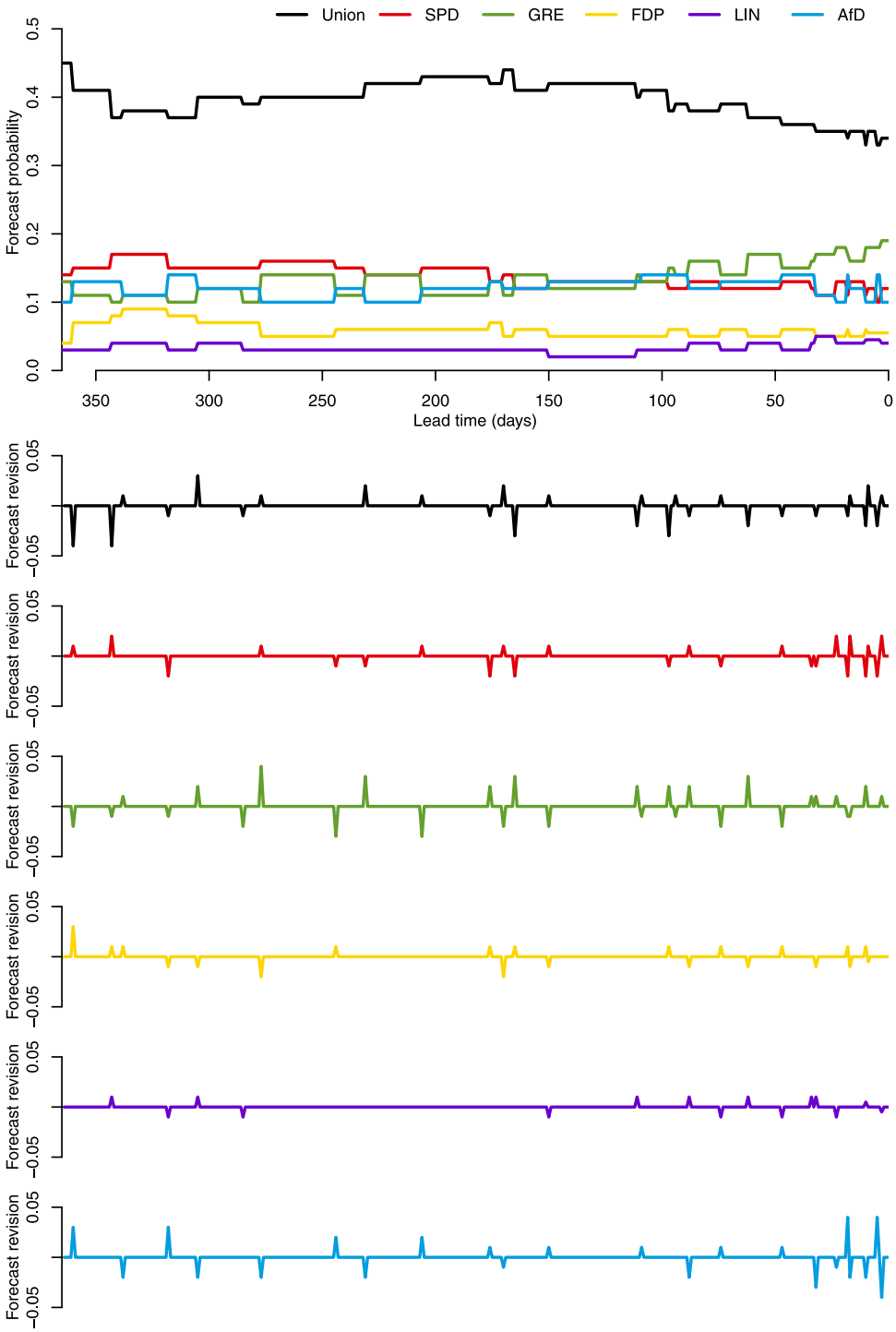


Fig. 3. Realized forecasts (top display) and forecast revisions (other displays) of (BY, 2018).

2000 to August 2023 (see Appendix, Tables A.8, A.9), where we estimated coefficients and standard errors for all forecasting systems which were based on at least four polls. Our empirical findings are consistent with results for one political election (and six forecasting systems) in the US (Regnier, 2017) and thus suggest that it is possible to improve the forecasts of poll-based multi-period fixed-event forecasting systems systematically based on past forecasts. Similar empirical evidence suggesting that forecasting systems are inefficient has been documented

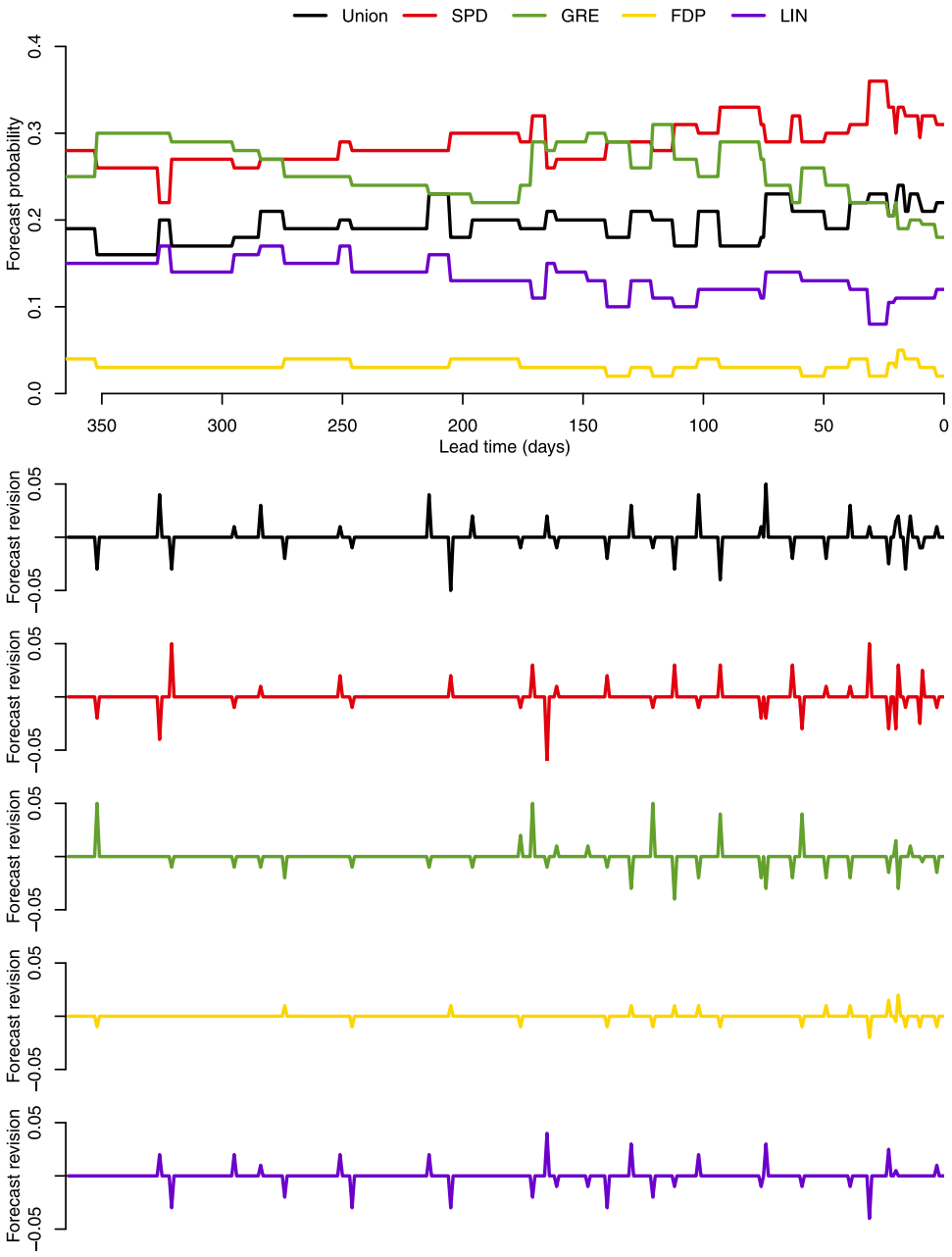


Fig. 4. Realized forecasts (top display) and forecast revisions (other displays) of (BE, 2011).

for a wide range of forecasting systems: Examples include forecasting systems generating oil price forecasts and forecasts for the energy consumption in the European countries (Nordhaus, 1987), gross-domestic product and inflation forecasts for the UK (Clements, 1997; Clements & Taylor, 2001) and the US (Clements et al., 2007) issued by various different institutions.

All our results up to this point indicate that poll-based forecasting systems for Germany are not L_2-weakly efficient. However, there are several explanations for our findings that merit consideration. Clements (1997) pointed out that potential over-differencing of stationary forecasts may lead to negative autocorrelations in the revisions. This may arise, when the process of forecasts X_t is uncorrelated and has zero mean conditional on all past forecasts. In this case, negative slope coefficients in the above test equations may still be consistent with weakly efficient forecasting systems. A referee pointed out that this is a matter of definition and some readers might object to considering a sequence of forecasts equal to a common value plus random noise as weakly efficient. The question of whether the above explanation is compatible

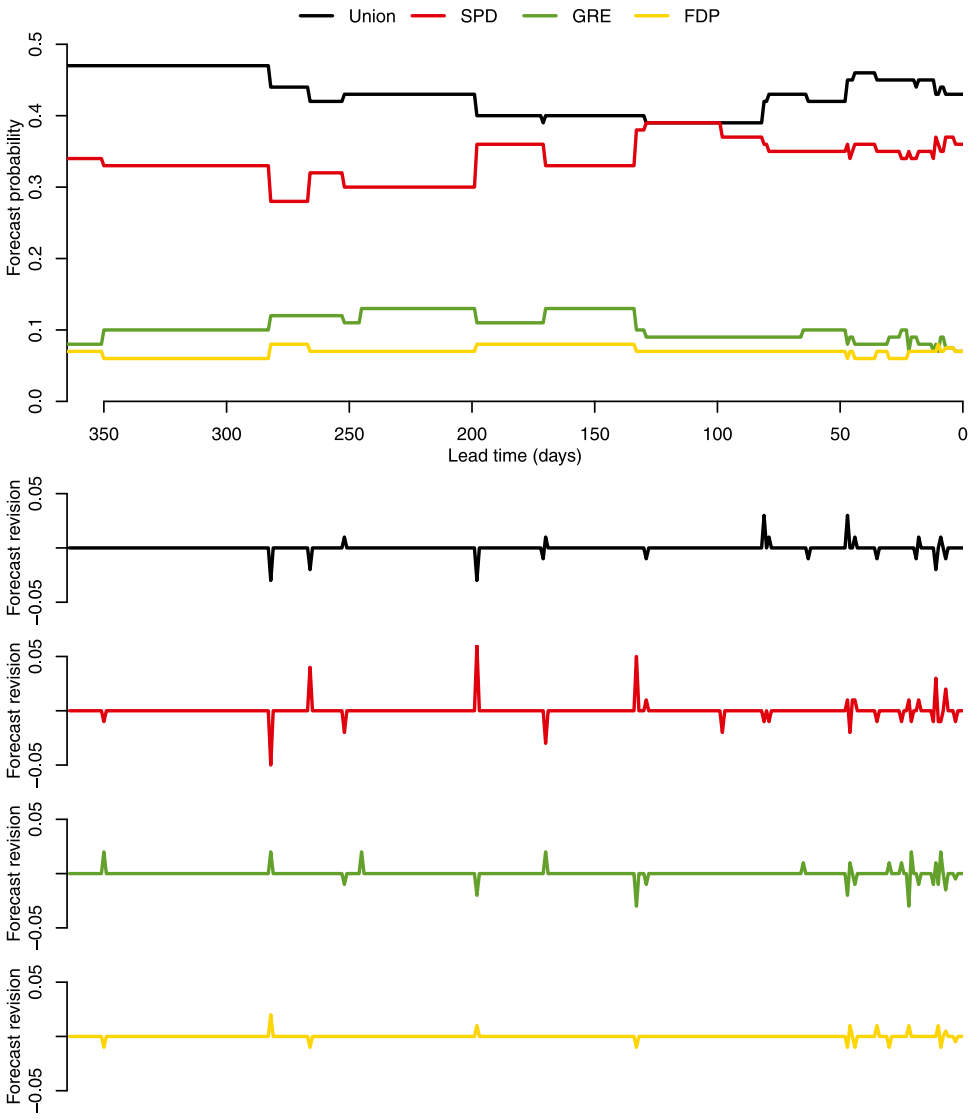


Fig. 5. Realized forecasts (top display) and forecast revisions (other displays) of (NW, 2005).

Table 7
Estimated forecast revision slope coefficients and standard errors for forecasting systems (BY, 2018), (BE, 2011), and (NW, 2005) based on test equations (18) with party-specific intercept and slope coefficient; (19) with party-specific intercept only; (20) when pooling across parties.

Party	(BY, 2018)		(BE, 2011)		(NW, 2005)	
	$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$
Eq. (18)						
Union	−0.348	0.127	−0.592	0.124	0.045	0.199
SPD	−0.615	0.178	−0.442	0.126	−0.320	0.121
GRE	−0.824	0.115	−0.212	0.134	−0.321	0.186
FDP	−0.247	0.214	−0.445	0.363	−0.540	0.367
LIN	−0.535	0.344	−0.497	0.163		
AfD	−0.449	0.124				
Eq. (19)	−0.538	0.062	−0.438	0.066	−0.262	0.088
Eq. (20)	−0.520	0.062	−0.433	0.066	−0.258	0.087

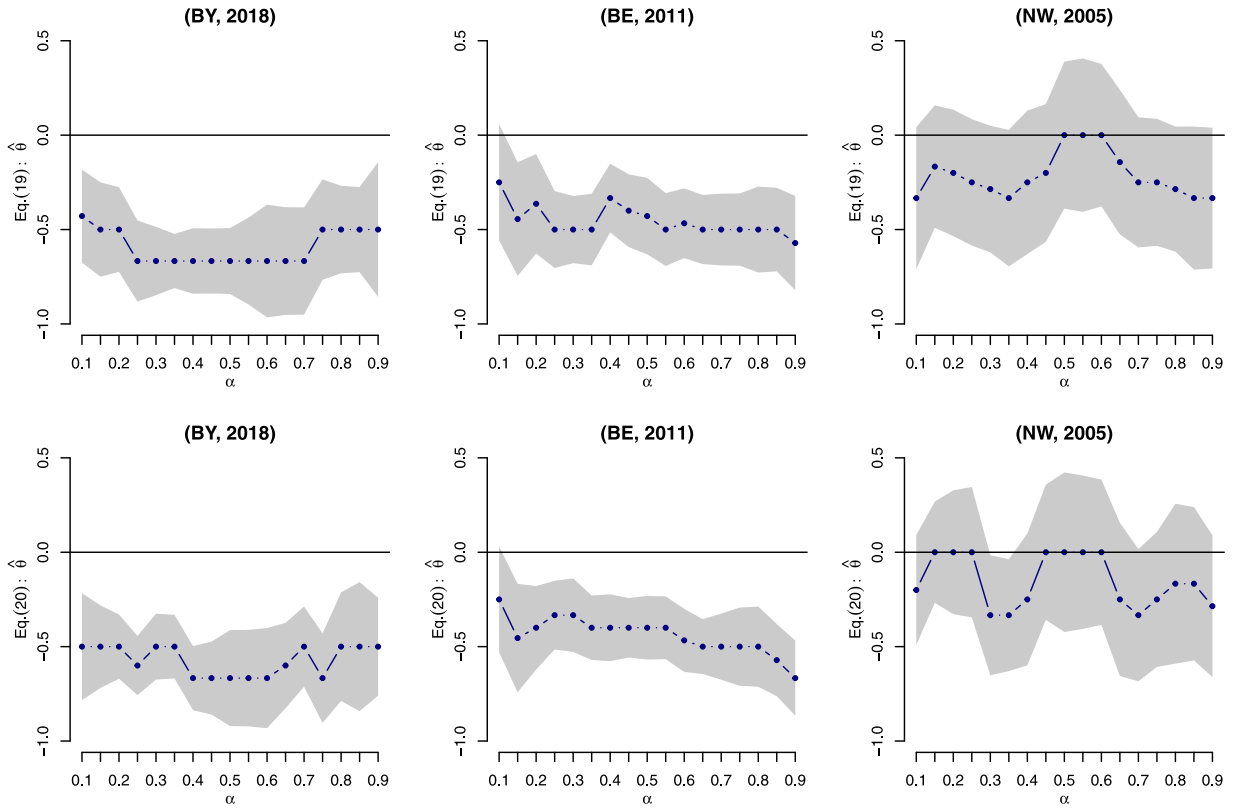


Fig. 6. Forecast revision slope coefficient and 95% pointwise confidence bands based on bootstraped standard errors from α-quantile regression estimation of Eqs. (19) in upper and (20) in lower displays for forecasting systems (BY, 2018), (BE, 2011), and (NW, 2005).

with the concept of weak efficiency remains for the reader to decide. In the context of election forecasting, stationary forecasts may result from the absence of information that affects the voting decision of the participants of the polls over the lead time. Another explanation is that the data used to investigate fixed event forecasting systems are typically limited in both, the cross-sectional and longitudinal dimension. In the context of election forecasting, this is especially the case as pooling data over elections is not feasible, as different elections represent different forecasting systems (Regnier, 2017). Further explanations are related to the sample analogue of the population orthogonality condition (9). In practice, the function $g(\cdot)$ is often assumed to be linear and only one lag is considered via (12) – while the function may be nonlinear, the dynamics may extend beyond the first lag, or both.

As discussed in Section 3, we address Clements (1997) explanation why Nordhaus' symmetric L_2-weak efficiency may be rejected for a given forecasting system: If the forecasting system was designed based on a loss function with asymmetrical weighting, while the evaluation for efficiency is based on least squares loss, the forecasting system may seem to overpredict systematically. Systematic overprediction indicates the presence of systematic bias, which is at odds with efficiency (Regnier, 2017). In Fig. 6 we display results for investigating the quantilegale difference hypothesis using condition (17). The upper (lower) displays show α-quantile regression estimates for the slope coefficient of test Eq. (19) (for test Eq. (20)) with pointwise 95% confidence bands based on bootstrapped standard errors. The results for the forecasting systems (BY, 2018) and (BE, 2011) illustrate that the α-quantilegale condition (17) is clearly separated from the null for all $\alpha \in \{0.10, 0.15, \ldots, 0.90\}$, while there is no evidence against L_α-weak efficiency for the forecasting system (NW, 2005). Quartile regression estimation results of Eq. (19) are shown in Tables A.10 and A.11 (see Appendix; further results for all $\alpha \in \{0.10, 0.15, \ldots, 0.90\}$ are available from the authors upon request). The forecasting systems (BW, 2011) and the illustrated (NW, 2005) are the only cases that include a relatively large number of polls and for which the empirical evidence does not indicate to reject the quartilegale difference hypothesis. Apart from these forecasting systems the hypothesis is called into question for all forecasting systems with more than 10 to 13 polls. The same limitations as pointed out above apply: Lags beyond the first may be relevant in condition (17).

In order to check whether forecasts become more informative with shorter lead times, all results in this section were also calculated (results are available upon request from the authors) for the last $\lambda_{\max} = 180$ days ahead of each state election (and the respective smaller sample sizes according to Table 4). The results are qualitatively identical to setting $\lambda_{\max} = 365$ and provide evidence against L_2- and L_α-weak efficiency.

6. Summary and outlook

We investigate the efficiency of election poll-based multi-period fixed-event probability-forecasting systems for German federal state elections. Weak efficiency of a forecasting system implies that its performance cannot be improved systematically by (functions of) past forecasts or forecast revisions. A forecasting system is L_2-weakly efficient if the forecasts (forecast revisions) evolve over time as martingales (martingale differences). We summarize the conditions and tests for weak efficiency of forecasting systems and then test the martingale difference null hypothesis – which is based on the least squares loss function – empirically for forecasting systems for all German federal state elections since the year 2000. We find that L_2-weak efficiency is rejected. Besides inefficiency of the forecasting systems in this sense, there are alternative explanations for this result: Data limitations, dynamic misspecification of the test equations, and wrong assumptions regarding the functional form of the orthogonality condition used to test for weak efficiency. Additionally, invoking a symmetric loss function such as least squares may not be appropriate. We account for the latter by empirically evaluating a quantilegale difference hypothesis, which is based on the asymmetrically weighted least absolute deviations loss function. From the existing literature on the efficiency of election polls, it is difficult to identify a specific choice for the asymmetry parameter. Therefore, we evaluate the hypothesis based on a grid of asymmetry parameters. Similar to the previous analyses of the martingale difference hypothesis, we find indications for L_α-inefficiency of the poll-based forecasting systems. Our empirical evidence on L_2-weak efficiency is in line with results from the literature on forecasting systems for senate elections in the US and suggests that poll-based multi-period fixed-event probability-forecasting systems for German federal state elections can be improved systematically based on efficient use of past information.

The empirical evidence against different weak efficiency concepts for poll-based forecasting systems suggests various directions for future research. Additionally, there is a number of alternatives for deriving the test equations and estimating their parameters.

Weak efficiency of a forecasting system rules out any information rigidities, which imply that new information is not incorporated into the system instantly. Vereda et al. (2021) provide two different hypotheses (and corresponding regression frameworks with structural interpretations) why information may only sink in gradually: Sticky information and noisy information. In the former case, forecasters face 'costs' such as a loss in reputation when revising their forecasts. This means that when new information becomes available, only a certain fraction of forecasters are willing to adjust their forecasts. In the latter case, new information can only be observed by the forecasters with a certain error. In the presence of information rigidities, it is possible to improve fixed-event forecasts systematically based on past information. This may be achieved by, for example, using functions of past forecasts (Pesaran & Pick, 2011; Pesaran et al., 2013). Alternatively, feedback about any systematic bias underlying the polls may also be incorporated to improve the forecasts generated by the forecasting system (Petropoulos et al., 2017).

There are further alternative scoring rules that can be employed to investigate the efficiency of forecasting systems beyond the central tendency of the empirical distribution. Quantilegale differences (Linton & Whang, 2007), for example, result from employing an asymmetrically weighted loss function based on least absolute deviations and provide a robust alternative to martingale differences by generalizing the central tendency to other areas of the conditional distribution of the forecast revisions given the full history of past forecast revisions. Similarly, an asymmetrically weighted version of squared error loss (or expectile loss; for a summary and comparison to quantile loss, see Kneib, 2013) can be used that yields expectilegale differences. Obtaining the asymmetry parameter is key in asymmetrically weighted least absolute- and least squares-based analysis. In the absence of knowledge about the parameter, suitable assumptions or methods for estmating the parameter are required. One alternative may be developing a structural dynamic quantile model for the voting decision of eligible voters, which is similar to the utility model of de Castro et al. (2020).

A further challenge in the evaluation of poll-based forecasting systems for politcal elections is the typically small number of events, and the correlation of forecastable outcomes such as the state-level results across the different parties and senate seats (Regnier, 2017). Discussing plausible sets of assumptions for settings in which no large datasets are available and developing data-driven procedures to account for the correlation structures and adding-up restrictions may also be worthwhile research endeavours.

Rejections of the martingale difference null hypothesis might also happen due to over-differencing of stationary forecasts. For example, over-differencing of white noise leads to negative autocorrelations in the forecast revisions of -0.5, which are likely to be interpreted as indications that the forecasting system is not efficient. If the innovation of the process of forecasts across adjacent time periods is white noise, however, negative coefficients may still be consistent with weak efficiency. In the context of election forecasting, stationary forecasts may result from the absence of information that affects the voting decision of the participants of the polls over the lead time (Clements, 1997).

Finally, as an alternative to OLS estimation of the test Eqs. (19) and (20), parameter estimates can be obtained from the generalized method of moments (GMM). The method resolves overidentification when the number of orthogonality conditions that is available for estimating the parameters of a given test equation exceeds the actual number of parameters. Additionally, endogeneity resulting from the unobservable party-specific effect (Davies & Lahiri, 1995) or systematic measurement error (Patton & Timmermann, 2011) can be resolved by GMM. Another setting where GMM is suitable is the overidentified setting sketched in de Castro et al. (2020), where the

Table A.8
Estimated forecast revision slope coefficients and standard errors for all forecasting systems (part 1).

State	Year	# Parties	# Polls	Eq. (19)		Eq. (20)	
				$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$
BB	2004	5	12	−0.439	0.133	−0.359	0.130
BB	2009	5	7	−0.590	0.168	−0.534	0.167
BB	2014	5	10	−0.302	0.151	−0.257	0.147
BB	2019	6	13	−0.527	0.101	−0.498	0.100
BE	2001	5	16	−0.609	0.092	−0.564	0.095
BE	2006	5	25	−0.465	0.075	−0.463	0.074
BE	2011	5	37	−0.438	0.066	−0.433	0.066
BE	2016	5	26	−0.695	0.066	−0.681	0.066
BE	2021	6	15	−0.371	0.101	−0.347	0.098
BE	2023	6	13	−0.603	0.094	−0.558	0.094
BW	2001	4	11	−0.371	0.135	−0.345	0.131
BW	2006	4	8	−0.363	0.167	−0.351	0.155
BW	2011	5	29	−0.062	0.086	−0.057	0.085
BW	2016	6	23	−0.413	0.079	−0.360	0.079
BW	2021	6	15	−0.650	0.063	−0.646	0.062
BY	2003	4	7	−0.484	0.151	−0.355	0.147
BY	2008	5	25	−0.266	0.091	−0.258	0.089
BY	2013	4	24	−0.372	0.100	−0.372	0.099
BY	2018	6	31	−0.538	0.062	−0.520	0.062
BY	2023	5	27	−0.479	0.075	−0.475	0.074
HB	2003	4	4	−0.565	0.313	−0.387	0.297
HB	2011	5	5	−0.278	0.227	0.074	0.199
HB	2015	6	5	−0.471	0.183	−0.439	0.167
HB	2019	6	9	−0.589	0.115	−0.522	0.117
HB	2023	5	7	−0.218	0.177	−0.091	0.161
HE	2003	4	10	−0.045	0.159	0.045	0.145
HE	2008	5	18	−0.464	0.099	−0.421	0.099
HE	2009	5	14	−0.328	0.108	−0.250	0.106

orthogonality conditions are nonlinear functionals. Yet another setting where GMM may be employed is provided by Patton and Timmermann (2007a). The approach carries out a data-driven approximation of the loss function based on a series of realizations of the target variable. Such realizations are, however, not available in fixed-event forecasting and it may be a further worthwhile research endeavour to investigate if the approach of Patton and Timmermann (2007a) can be extended to the context of fixed-event forecasting.

CRediT authorship contribution statement

Markus Fritsch: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Harry Haupt:** Conceptualization, Formal analysis, Investigation, Methodology, Writing – original draft, Writing – review & editing. **Joachim Schnurbus:** Conceptualization, Investigation, Software, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

All data and code required to reproduce the results in this paper is available from: https://github.com/markusfritsch/fixedEventFC.

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Appendix. Results for all state elections

Tables A.8 and A.9 show the slope coefficient estimates and the corresponding standard errors for evaluating the martingale difference hypothesis (11) for the poll-based forecasting systems for all state elections since the year 2000. Estimates were obtained based on the forecast revisions z_t for all systems which are based on at least four polls conducted within 365 days ahead of the state election. Note that the number of revisions equals the number of available polls minus one. Equivalent results for the forecasts x_t are available upon request from the authors.

Tables A.10 and A.11 show the slope coefficient estimates and the corresponding standard errors for evaluating the $\alpha = \{0.25, 0.5, 0.75\}$-quantilegale difference hypothesis (16) for the poll-based forecasting systems for all state elections since the year 2000. Estimates were obtained based on the forecast revisions z_t for all systems which are based on at least four polls conducted within 365 days ahead of the state election. Note that the number of revisions equals the number of available polls minus one. Equivalent results for the forecasts x_t are available upon request from the authors.

Table A.9
Estimated forecast revision slope coefficients and standard errors for all forecasting systems (part 2).

State	Year	# Parties	# Polls	Eq. (19)		Eq. (20)	
				$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$
HE	2013	5	10	−0.598	0.131	−0.489	0.128
HE	2018	6	15	−0.419	0.103	−0.375	0.101
HE	2023	6	12	−0.153	0.116	−0.063	0.111
HH	2001	4	14	−0.398	0.131	−0.393	0.127
HH	2004	4	24	−0.187	0.084	−0.187	0.083
HH	2008	5	23	−0.422	0.088	−0.414	0.087
HH	2011	5	19	−0.429	0.093	−0.405	0.092
HH	2015	5	12	−0.500	0.142	−0.492	0.135
HH	2020	6	18	−0.673	0.083	−0.661	0.082
MV	2002	5	17	−0.350	0.109	−0.333	0.106
MV	2006	5	13	−0.346	0.089	−0.336	0.086
MV	2011	5	11	−0.242	0.101	−0.201	0.094
MV	2016	5	10	−0.037	0.152	0.102	0.141
MV	2021	6	14	−0.228	0.115	−0.122	0.112
NI	2003	4	18	−0.523	0.099	−0.506	0.097
NI	2008	5	14	−0.594	0.095	−0.587	0.093
NI	2013	5	19	−0.401	0.098	−0.389	0.096
NI	2017	6	14	−0.358	0.107	−0.350	0.104
NI	2022	6	19	−0.219	0.090	−0.214	0.088
NW	2005	4	32	−0.262	0.088	−0.258	0.087
NW	2010	5	28	−0.593	0.066	−0.579	0.066
NW	2012	5	28	−0.394	0.080	−0.372	0.079
NW	2017	6	21	−0.391	0.088	−0.384	0.086
NW	2022	6	21	−0.388	0.087	−0.367	0.086
RP	2001	4	12	−0.464	0.135	−0.461	0.131
RP	2006	4	13	−0.497	0.129	−0.478	0.127
RP	2011	5	22	−0.263	0.092	−0.258	0.090
RP	2016	6	21	−0.528	0.081	−0.479	0.081
RP	2021	6	15	−0.382	0.101	−0.337	0.100
SH	2005	4	13	−0.610	0.118	−0.509	0.121
SH	2009	5	13	−0.541	0.118	−0.474	0.118
SH	2012	5	14	−0.349	0.106	−0.325	0.106
SH	2017	6	9	−0.590	0.113	−0.558	0.109
SH	2022	5	13	−0.674	0.092	−0.662	0.092
SL	2004	4	8	−0.448	0.191	−0.401	0.184
SL	2009	5	8	−0.308	0.147	−0.265	0.146
SL	2012	5	7	−0.228	0.169	−0.202	0.162
SL	2017	6	10	−0.112	0.112	−0.117	0.107
SL	2022	6	9	−0.453	0.107	−0.249	0.123
SN	2004	5	16	−0.216	0.111	−0.192	0.109
SN	2009	5	14	−0.415	0.098	−0.399	0.095
SN	2014	5	13	−0.471	0.103	−0.439	0.102
SN	2019	6	15	−0.392	0.106	−0.381	0.103
ST	2002	5	9	−0.392	0.147	−0.331	0.141
ST	2006	5	12	−0.270	0.122	−0.269	0.118
ST	2011	5	7	−0.245	0.152	−0.193	0.142
ST	2016	4	13	−0.331	0.144	−0.321	0.139
ST	2021	6	10	−0.179	0.147	−0.120	0.143
TH	2004	5	17	−0.496	0.097	−0.488	0.095
TH	2009	5	15	−0.445	0.112	−0.417	0.111
TH	2014	4	12	−0.175	0.145	−0.167	0.139
TH	2019	6	14	−0.436	0.088	−0.423	0.087

Table A.10
Forecast revision slope coefficients and standard errors based on α-quantile regression of Eq. (19) for all forecasting systems (part 1).

State	Year	# Parties	# Polls	$\alpha = 0.25$		$\alpha = 0.50$		$\alpha = 0.75$	
				$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$
BB	2004	5	12	−0.273	0.290	−0.600	0.205	−0.667	0.193
BB	2009	5	7	−0.500	0.391	−0.333	0.312	−0.500	0.297
BB	2014	5	10	−0.143	0.220	−0.364	0.215	−0.400	0.280
BB	2019	6	13	−0.667	0.156	−0.500	0.141	−0.500	0.164
BE	2001	5	16	−0.800	0.190	−0.750	0.121	−0.667	0.148
BE	2006	5	25	−0.500	0.106	−0.500	0.159	−0.667	0.177
BE	2011	5	37	−0.500	0.105	−0.429	0.097	−0.500	0.098
BE	2016	5	26	−0.750	0.118	−0.909	0.120	−0.667	0.108
BE	2021	6	15	−0.500	0.134	−0.455	0.143	−0.500	0.186
BE	2023	6	13	−0.500	0.227	−0.667	0.221	−0.714	0.167
BW	2001	4	11	−0.333	0.272	−0.333	0.182	−0.333	0.128
BW	2006	4	8	−0.375	0.300	0.000	0.253	−0.400	0.295
BW	2011	5	29	−0.100	0.144	−0.109	0.148	−0.200	0.167
BW	2016	6	23	−0.500	0.109	−0.500	0.099	−0.500	0.135
BW	2021	6	15	−0.778	0.152	−0.778	0.134	−0.500	0.194
BY	2003	4	7	−0.250	0.264	−0.667	0.280	−0.400	0.321
BY	2008	5	25	−0.333	0.163	−0.250	0.205	−0.333	0.277
BY	2013	4	24	−0.333	0.202	−0.167	0.175	−0.500	0.216
BY	2018	6	31	−0.667	0.091	−0.667	0.100	−0.500	0.138
BY	2023	5	27	−0.500	0.122	−0.500	0.114	−0.500	0.112
HB	2003	4	4	−0.333	1.037	−0.333	1.146	−0.667	1.211
HB	2011	5	5	0.000	0.380	−0.292	0.366	−0.500	0.416
HB	2015	6	5	−0.750	0.393	−0.500	0.279	−0.625	0.312
HB	2019	6	9	−0.500	0.178	−0.643	0.224	−0.600	0.288
HB	2023	5	7	−0.250	0.271	0.000	0.269	−0.250	0.358
HE	2003	4	10	−0.333	0.311	0.000	0.207	0.000	0.254
HE	2008	5	18	−0.500	0.180	−0.500	0.085	−0.500	0.134
HE	2009	5	14	−0.400	0.213	−0.333	0.215	−0.500	0.223
HE	2013	5	10	−0.500	0.233	−0.500	0.237	−0.667	0.247
HE	2018	6	15	−0.400	0.197	−0.400	0.160	−0.333	0.169
HE	2023	6	12	−0.250	0.181	0.000	0.247	0.000	0.197
HH	2001	4	14	−0.333	0.183	−0.400	0.162	−0.500	0.231
HH	2004	4	24	−0.333	0.241	−0.333	0.198	−0.333	0.214
HH	2008	5	23	−0.400	0.153	−0.500	0.141	−0.500	0.148
HH	2011	5	19	−0.556	0.097	−0.500	0.141	−0.333	0.209
HH	2015	5	12	−0.213	0.308	−0.462	0.233	−0.526	0.315
HH	2020	6	18	−0.556	0.160	−0.667	0.166	−0.800	0.184
MV	2002	5	17	−0.333	0.152	−0.375	0.149	−0.500	0.183
MV	2006	5	13	−0.375	0.163	−0.286	0.136	−0.200	0.173
MV	2011	5	11	−0.455	0.153	−0.308	0.191	−0.125	0.221
MV	2016	5	10	0.000	0.252	−0.021	0.246	0.000	0.278
MV	2021	6	14	0.000	0.214	−0.200	0.176	−0.200	0.176
NI	2003	4	18	−0.500	0.150	−0.500	0.117	−0.385	0.132
NI	2008	5	14	−0.667	0.268	−0.500	0.259	−0.667	0.328
NI	2013	5	19	−0.500	0.136	−0.333	0.147	−0.556	0.202
NI	2017	6	14	−0.200	0.198	−0.333	0.148	−0.250	0.168
NI	2022	6	19	−0.400	0.176	−0.250	0.217	−0.375	0.213
NW	2005	4	32	−0.250	0.155	0.000	0.195	−0.250	0.188
NW	2010	5	28	−0.500	0.108	−0.500	0.091	−0.636	0.091
NW	2012	5	28	−0.500	0.153	−0.400	0.135	−0.333	0.182
NW	2017	6	21	−0.500	0.117	−0.333	0.163	−0.100	0.185
NW	2022	6	21	−0.455	0.084	−0.500	0.137	−0.333	0.147
RP	2001	4	12	−0.400	0.158	−0.500	0.148	−0.333	0.232
RP	2006	4	13	−0.250	0.209	−0.500	0.271	−0.500	0.216
RP	2011	5	22	−0.333	0.161	−0.200	0.136	−0.083	0.170
RP	2016	6	21	−0.500	0.136	−0.667	0.136	−0.500	0.131
RP	2021	6	15	−0.429	0.127	−0.500	0.098	−0.500	0.189
SH	2005	4	13	−0.500	0.224	−0.500	0.235	−0.500	0.280
SH	2009	5	13	−0.375	0.185	−0.400	0.214	−0.600	0.259
SH	2012	5	14	−0.500	0.176	−0.500	0.249	−0.333	0.173
SH	2017	6	9	−0.500	0.256	−0.286	0.317	−0.200	0.373
SH	2022	5	13	−0.500	0.170	−0.667	0.180	−0.667	0.186
SL	2004	4	8	−0.300	0.356	−0.500	0.271	−0.596	0.191
SL	2009	5	8	−0.333	0.166	−0.200	0.238	−0.333	0.294
SL	2012	5	7	−0.333	0.266	−0.200	0.221	0.000	0.282
SL	2017	6	10	0.000	0.173	−0.333	0.168	−0.143	0.153
SL	2022	6	9	−0.556	0.223	−0.300	0.253	−0.500	0.221

Table A.11
Forecast revision slope coefficients and standard errors based on α-quantile regression of Eq. (19) for all forecasting systems (part 2).

State	Year	# Parties	# Polls	$\alpha = 0.25$		$\alpha = 0.50$		$\alpha = 0.75$	
				$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$	$\widehat{\theta}$	$s.e.(\widehat{\theta})$
SN	2004	5	16	−0.333	0.254	−0.200	0.188	−0.250	0.159
SN	2009	5	14	−0.333	0.189	−0.571	0.201	−0.500	0.180
SN	2014	5	13	−0.500	0.151	−0.600	0.177	−0.500	0.231
SN	2019	6	15	−0.333	0.230	−0.333	0.201	−0.444	0.258
ST	2002	5	9	−0.500	0.209	−0.333	0.207	−0.200	0.296
ST	2006	5	12	−0.567	0.348	−0.548	0.191	−0.417	0.154
ST	2011	5	7	−0.167	0.166	−0.200	0.212	−0.500	0.315
ST	2016	4	13	−0.333	0.223	−0.500	0.213	−0.417	0.284
ST	2021	6	10	−0.400	0.256	−0.143	0.193	−0.125	0.170
TH	2004	5	17	−0.571	0.199	−0.600	0.139	−0.625	0.198
TH	2009	5	15	−0.333	0.242	−0.444	0.208	−0.500	0.152
TH	2014	4	12	−0.286	0.368	−0.333	0.242	−0.333	0.243
TH	2019	6	14	−0.500	0.112	−0.500	0.131	−0.500	0.132

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